Midterm Exam Solutions

Introduction to Statistics (통계의 이해) **2026 2nd Semester**

Year(학년): ___ Section(교반): ___ Cadet Number(교번): _____ Name(성명): _____ Score: _____

- **All solutions must include a detailed step-by-step explanation.**
- **If an answer has five or more decimal places, round to the fourth decimal place.**
- **Use the reference materials provided on the last page of the exam.**

1. Write O if each statement is true and X if it is false. [7 points]

(1) Adding 5 to every observation increases the mean by 5 but does not change the variance. ( )

(2) A probability density function $f(x)$ always satisfies $0 \leq f(x) \leq 1$. ( )

(3) If $f(x, y)$ is the joint probability mass function of X and Y, then the marginal probability mass function of X is $f_X(x) = \sum_{\forall y} f(x, y)$. ( )

(4) If $X \sim B(1, p)$, then $E(X) = p$ and $\mathrm{Var}(X) = p(1-p)$. ( )

(5) If $X \sim N(\mu, \sigma^2)$, then $\dfrac{X - \mu}{\sigma} \sim N(0, 1)$. ( )

(6) By the Central Limit Theorem, the population distribution becomes approximately normal as the sample size increases. ( )

(7) An unbiased estimator gives an estimate equal to the true value of the parameter. ( )

Solution: (1) O / (2) X / (3) O / (4) O / (5) O / (6) X / (7) X

(2) A probability density function only needs $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$; its value can be larger than 1.

(6) The Central Limit Theorem is about the sample mean $\bar{X}$: when n is large, $\bar{X}$ is approximately normal, regardless of the population distribution. The population distribution itself does not change.

(7) An estimator $\hat{\theta}$ is unbiased if $E(\hat{\theta}) = \theta$ (correct on average); a single estimate does not have to equal θ.

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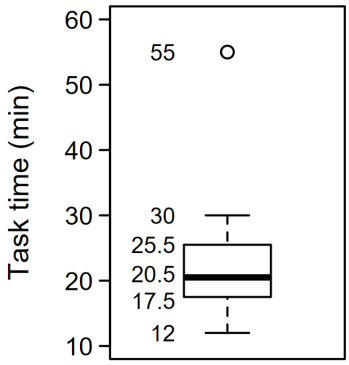
2. The plots below show the task times (in minutes) of 20 cadets. Fill in each blank or choose the correct answer in the parentheses. [10 points]

Task times of 20 cadets

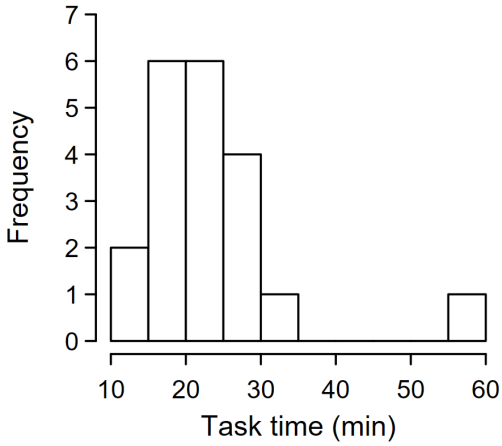
Stem-and-leaf plot

Stem	Leaf
1	2 4 5 6 7 8 8 9
2	0 0 1 2 3 4 5 6 7 8
3	0
4	
5	5

Box plot



Histogram



(Stem-and-leaf key: 1 | 2 means 12 minutes.)

(1) The number of cadets who took exactly 20 minutes is ( ).

(2) The median is ( ) minutes.

(3) The interquartile range (IQR) is ( ) minutes.

(4) If the bin width changes from 5 to 10 minutes, the histogram (can look different / must look the same), and the mean of the data (changes / does not change).

(5) The outlier 55 was a recording mistake, and the correct value is 40. After this correction, the median (increases / does not change / decreases), and the mean (increases / does not change / decreases).

Solution: (1) 2 / (2) 20.5 / (3) 8 / (4) can look different, does not change / (5) does not change, decreases

(1) Stem 2 has two leaves 0, so the value 20 appears twice.

(3) IQR = Q3 − Q1 = 25.5 − 17.5 = 8 (Q1 and Q3 are read from the box plot).

(4) Changing the bins changes how the data are grouped, so the histogram can look different. The data values themselves do not change, so the mean does not change.

(5) 40 is still larger than the two middle values (20 and 21), so the median stays 20.5. The total decreases by 15, so the mean decreases by $15/20 = 0.75$.

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3. The probability that an aircraft is in a radar's watch area is 0.2. If an aircraft is in the area, the radar sounds an alarm with probability 0.9. If not, it still sounds an alarm with probability 0.15. Answer the following questions. [10 points]

(1) Find the probability that the radar sounds an alarm. (Define the events first.)

Solution: Let A be the event that an aircraft is in the watch area, and B be the event that the radar sounds an alarm. Then

$$P(A) = 0.2, \quad P(A^c) = 0.8, \quad P(B \mid A) = 0.9, \quad P(B \mid A^c) = 0.15.$$

Since A and A^c form a partition of the sample space, the probability that the radar sounds an alarm is

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) \\ &= P(A)P(B \mid A) + P(A^c)P(B \mid A^c) \\ &= 0.2 \times 0.9 + 0.8 \times 0.15 \\ &= 0.3. \end{aligned}$$

(2) If the radar sounds an alarm, find the probability that an aircraft is really in the area.

Solution: Since A and A^c form a partition of the sample space, by Bayes' theorem,

$$\begin{aligned} P(A \mid B) &= \frac{P(A)P(B \mid A)}{P(A)P(B \mid A) + P(A^c)P(B \mid A^c)} \\ &= \frac{0.2 \times 0.9}{0.2 \times 0.9 + 0.8 \times 0.15} \\ &= 0.6. \end{aligned}$$

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4. The joint probability density function of X and Y is

$$f(x,y) = \begin{cases} c\,(1+x^2 y), & -1 < x < 1,\ -1 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Answer the following questions. [15 points]

(1) Find the constant c.

Solution: The integral of a joint probability density function over the whole region must be 1. Integrating with respect to y first,

$$1 = \int_{-1}^{1}\int_{-1}^{1} c\,(1+x^2 y)\,dy\,dx = c\int_{-1}^{1}\left[\,y + \frac{x^2 y^2}{2}\,\right]_{y=-1}^{1} dx = c\int_{-1}^{1} 2\,dx = 4c \quad \Rightarrow \quad c = \frac{1}{4}.$$

(2) Find the marginal probability density functions of X and Y.

Solution: To find $f_X(x)$, integrate with respect to y; to find $f_Y(y)$, integrate with respect to x:

$$f_X(x) = \int_{-1}^{1} \frac{1}{4}(1+x^2 y)\,dy = \frac{1}{4}\left[\,y + \frac{x^2 y^2}{2}\,\right]_{y=-1}^{1} = \frac{1}{2}, \qquad -1 < x < 1,$$

$$f_Y(y) = \int_{-1}^{1} \frac{1}{4}(1+x^2 y)\,dx = \frac{1}{4}\left[\,x + \frac{x^3 y}{3}\,\right]_{x=-1}^{1} = \frac{1}{2} + \frac{y}{6}, \qquad -1 < y < 1,$$

and both are 0 otherwise.

(3) Find $E(X)$, $E(Y)$, and $E(XY)$.

Solution:

$$E(X) = \int_{-1}^{1} x\, f_X(x)\,dx = \frac{1}{2}\int_{-1}^{1} x\,dx = 0,$$

$$E(Y) = \int_{-1}^{1} y\, f_Y(y)\,dy = \frac{1}{2}\int_{-1}^{1} y\,dy + \frac{1}{6}\int_{-1}^{1} y^2\,dy = 0 + \frac{1}{6}\times\frac{2}{3} = \frac{1}{9},$$

$$E(XY) = \int_{-1}^{1}\int_{-1}^{1} xy\, f(x,y)\,dy\,dx = \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1} (xy + x^3 y^2)\,dy\,dx = \frac{1}{4}\int_{-1}^{1} \frac{2x^3}{3}\,dx = 0.$$

(4) Find $\mathrm{Cov}(X,Y)$.

Solution: $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0 - 0 \times \dfrac{1}{9} = 0.$

(5) Determine whether X and Y are independent.

Solution: From (1) and (2), for $-1 < x < 1$ and $-1 < y < 1$,

$$f(x,y) = \frac{1}{4}(1+x^2 y), \qquad f_X(x) f_Y(y) = \frac{1}{2}\left(\frac{1}{2} + \frac{y}{6}\right) = \frac{1}{4} + \frac{y}{12}.$$

Since $f(x,y) \neq f_X(x) f_Y(y)$ (for example, $f\left(0, \tfrac{1}{2}\right) = \tfrac{1}{4}$ but $f_X(0) f_Y\left(\tfrac{1}{2}\right) = \tfrac{7}{24}$), X and Y are **not independent**. (Note: $\mathrm{Cov}(X,Y) = 0$ even though X and Y are not independent.)

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5. The weights (in grams) of products from Company A and Company B follow normal distributions with means $\mu_A = 500$, $\mu_B = 480$ and standard deviations $\sigma_A = 20$, $\sigma_B = 15$. One product is chosen independently from each company. Find the probability that the Company A product is at least 45 g heavier than the Company B product. (Define the random variables first.) [8 points]

Solution: Let X be the weight of the product from Company A, and Y be the weight of the product from Company B. Then

$$X \sim N(500,\ 20^2), \qquad Y \sim N(480,\ 15^2).$$

Since X and Y are independent, by the properties of the normal distribution,

$$X - Y \sim N(500 - 480,\ 20^2 + 15^2) = N(20,\ 25^2).$$

Therefore,

$$P(X \geq Y + 45) = P(X - Y \geq 45) = P\left(\frac{X - Y - 20}{25} \geq \frac{45 - 20}{25} \right)$$
$$= P(Z \geq 1) = 1 - P(Z \leq 1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587.$$

6. The scores on a test follow a normal distribution with mean $\mu = 72$ points and standard deviation $\sigma = 9$ points. The top 5% of students get an A. Find the cutoff score for an A. (Define the random variable first.) [8 points]

Solution: Let X be the test score. Then $X \sim N(72,\ 9^2)$.

To find the cutoff for the top 5%, we find x such that $P(X \geq x) = 0.05$. First, the value z with $P(Z \geq z) = 0.05$ is $z_{0.05} = 1.6449$, that is, $P(Z \geq 1.6449) = 0.05$. Therefore,

$$0.05 = P(X \geq x) = P\left(\frac{X - 72}{9} \geq \frac{x - 72}{9} \right) = P(Z \geq 1.6449)$$

$$\Rightarrow\ \frac{x - 72}{9} = 1.6449\ \Rightarrow\ x = 72 + 1.6449 \times 9 = 86.8041.$$

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7. The lifetime X (in 1,000 hours) of a military radar part has the probability density function

$$f(x) = \begin{cases} \dfrac{1}{2}e^{-x/2}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Answer the following questions. [15 points]

(1) Show that $E(X) = 2$. (Note: $E(X^2) = 8$.)

Solution:

$$E(X) = \int_0^\infty x\, f(x)\, dx = \int_0^\infty \frac{x}{2}\, e^{-x/2}\, dx.$$

For integration by parts, let $u = x$ and $dv = \frac{1}{2}e^{-x/2}\, dx$. Then $du = dx$ and $v = -e^{-x/2}$. Therefore,

$$E(X) = \Big[-xe^{-x/2} \Big]_0^\infty + \int_0^\infty e^{-x/2}\, dx = \Big[-xe^{-x/2} \Big]_0^\infty + \Big[-2e^{-x/2} \Big]_0^\infty = 0 + \big(0 - (-2)\big) = 2.$$

(2) For a random sample $X_1, \ldots, X_{64}$ of $n = 64$ parts, find the approximate distribution of the sample mean $\bar{X}$, and name the theorem you used.

Solution: From (1), the population mean is $E(X) = 2$. Since $E(X^2) = 8$, the population variance is

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 8 - 2^2 = 4.$$

So the mean and variance of the sample mean are

$$E(\bar{X}) = E(X) = 2, \qquad \mathrm{Var}(\bar{X}) = \frac{\mathrm{Var}(X)}{n} = \frac{4}{64} = \frac{1}{16}.$$

Since the sample size $n = 64$ is large enough, by the **Central Limit Theorem**, the sample mean approximately follows a normal distribution:

$$\bar{X} \mathrel{\dot\sim} N\left(2,\ \frac{1}{16}\right), \quad \text{or, after standardizing,} \quad Z = \frac{\bar{X} - 2}{1/4} \mathrel{\dot\sim} N(0, 1).$$

(3) Find the approximate value of $P(1.5 \le \bar{X} \le 2.5)$.

Solution:

$$P(1.5 \le \bar{X} \le 2.5) = P\left(\frac{1.5 - 2}{1/4} \le \frac{\bar{X} - 2}{1/4} \le \frac{2.5 - 2}{1/4} \right) \approx P(-2 \le Z \le 2)$$
$$= \Phi(2) - \{1 - \Phi(2)\} = 0.9772 - 0.0228 = 0.9544.$$

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8. The number of defects X on a piece of military equipment has the following probability distribution.

x	0	1	2
$P(X=x)$	5/8	2/8	1/8

Let $\bar{X}_n$ be the sample mean of a random sample $X_1, \ldots, X_n$. Answer the following questions. [12 points]

(1) For $n=2$, find the probability distribution of $\bar{X}_2$. Then find $P(\bar{X}_2 \geq 0.8)$.

Solution: Since X_1, X_2 is a random sample, the joint probability distribution $P(X_1=x_1, X_2=x_2) = P(X_1=x_1)P(X_2=x_2)$ is

$x_1 \setminus x_2$	**0**	**1**	**2**
0	25/64	10/64	5/64
1	10/64	4/64	2/64
2	5/64	2/64	1/64

and the probability distribution of the sample mean $\bar{X}_2 = (X_1+X_2)/2$ is

$\bar{x}$	0	0.5	1	1.5	2
$P(\bar{X}_2=\bar{x})$	25/64	20/64	14/64	4/64	1/64

Therefore,

$$P(\bar{X}_2 \geq 0.8) = P(\bar{X}_2=1) + P(\bar{X}_2=1.5) + P(\bar{X}_2=2) = \frac{14}{64} + \frac{4}{64} + \frac{1}{64} = \frac{19}{64} \approx 0.2969.$$

(2) For $n=50$, use the Central Limit Theorem to find the approximate distribution of $\bar{X}_{50}$. Then find the approximate value of $P(\bar{X}_{50} \geq 0.8)$.

Solution:

$$E(X) = 0\times\frac{5}{8} + 1\times\frac{2}{8} + 2\times\frac{1}{8} = \frac{1}{2}, \qquad E(X^2) = 0^2\times\frac{5}{8} + 1^2\times\frac{2}{8} + 2^2\times\frac{1}{8} = \frac{3}{4},$$

so $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$. Therefore, the mean and variance of the sample mean are

$$E(\bar{X}_{50}) = E(X) = \frac{1}{2}, \qquad \mathrm{Var}(\bar{X}_{50}) = \frac{\mathrm{Var}(X)}{50} = \frac{1}{2}\times\frac{1}{50} = \frac{1}{100}.$$

Since the sample size $n=50$ is large enough, by the Central Limit Theorem,

$$\bar{X}_{50} \dot{\sim} N(0.5,\ 0.01), \quad \text{or, after standardizing,} \quad Z = \frac{\bar{X}_{50} - 0.5}{0.1} \dot{\sim} N(0,1).$$

Therefore,

$$P(\bar{X}_{50} \geq 0.8) = P\left(\frac{\bar{X}_{50}-0.5}{0.1} \geq \frac{0.8-0.5}{0.1}\right) \approx P(Z \geq 3) = 1 - \Phi(3) = 1 - 0.9987 = 0.0013.$$

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9. Let $X_1, X_2, \ldots, X_n$ be a random sample from the uniform distribution on the interval $(\theta,\, \theta+1)$. Answer the following questions. [15 points]

(1) Find $E(X_1)$ and $\mathrm{Var}(X_1)$.

Solution: Since X_1 has a uniform distribution on the interval $(\theta,\, \theta+1)$,

$$E(X_1) = \frac{\theta + (\theta+1)}{2} = \theta + \frac{1}{2}, \qquad \mathrm{Var}(X_1) = \frac{\{(\theta+1) - \theta\}^2}{12} = \frac{1}{12}.$$

(2) Find $E(\bar{X})$ and $\mathrm{Var}(\bar{X})$, where $\bar{X}$ is the sample mean.

Solution: The expected value of the sample mean equals the population mean, and the variance of the sample mean is the population variance divided by n:

$$E(\bar{X}) = E(X_1) = \theta + \frac{1}{2}, \qquad \mathrm{Var}(\bar{X}) = \frac{\mathrm{Var}(X_1)}{n} = \frac{1}{12n}.$$

(3) Using $\bar{X}$, find an estimator $\hat{\theta}$ of θ that is unbiased and consistent. Show both properties in detail.

Solution: From (2), $E(\bar{X}) = \theta + \frac{1}{2}$. Subtracting $\frac{1}{2}$ from both sides gives $E\left(\bar{X} - \frac{1}{2}\right) = \theta$. So if we let

$$\hat{\theta} = \bar{X} - \frac{1}{2},$$

then $E(\hat{\theta}) = \theta$, and $\hat{\theta}$ is an **unbiased** estimator of θ.

Also, $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}\left(\bar{X} - \frac{1}{2}\right) = \mathrm{Var}(\bar{X}) = \dfrac{1}{12n}$, so

$$\lim_{n\to\infty} E(\hat{\theta}) = \lim_{n\to\infty} \theta = \theta, \qquad \lim_{n\to\infty} \mathrm{Var}(\hat{\theta}) = \lim_{n\to\infty} \frac{1}{12n} = 0.$$

Therefore, $\hat{\theta} = \bar{X} - \frac{1}{2}$ is also a **consistent** estimator of θ.

※ **Reference Table** Let $Z \sim N(0,1)$ and $\Phi(z) = P(Z \le z)$.

$\Phi(0.25) = 0.5987$	$\Phi(0.5) = 0.6915$	$\Phi(0.75) = 0.7734$	$\Phi(1) = 0.8413$
$\Phi(1.25) = 0.8944$	$\Phi(1.5) = 0.9332$	$\Phi(1.75) = 0.9599$	$\Phi(2) = 0.9772$
$\Phi(2.25) = 0.9878$	$\Phi(2.5) = 0.9938$	$\Phi(2.75) = 0.9970$	$\Phi(3) = 0.9987$
$z_{0.30} = 0.5244$	$z_{0.20} = 0.8416$	$z_{0.15} = 1.0364$	$z_{0.10} = 1.2816$
$z_{0.05} = 1.6449$	$z_{0.025} = 1.9600$	$z_{0.01} = 2.3263$	$z_{0.005} = 2.5758$
$\chi^2_{0.05}(7) = 14.067$	$\chi^2_{0.025}(7) = 16.013$	$\chi^2_{0.05}(8) = 15.507$	$\chi^2_{0.025}(8) = 17.535$
$\chi^2_{0.05}(9) = 16.919$	$\chi^2_{0.025}(9) = 19.023$	$\chi^2_{0.05}(10) = 18.307$	$\chi^2_{0.025}(10) = 20.483$